Evidence Budgeting for Edge AI under Degraded Connectivity: Information-Matched Validation of Adaptivity, Admissibility and Adversarial Metadata on Synthetic and Real Disaster Data

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Abstract*—Edge AI on degraded field networks must decide what evidence to move and when to stop waiting. We formulate this as evidence budgeting across concurrent incident windows: a multiple-choice knapsack over a shared byte budget; the knapsack's latency term prices transfer time together with the forecast delay of late evidence, so that waiting becomes an explicit decision with a closed-form threshold. Evidence is fused only when its location, time and hazard tag are admissible; each decision carries a conformal set and a provenance record. We compare the formulation against budget-matched simple rules that receive the same information, across nine synthetic conditions at twenty seeds and a reduced CrisisMMD workload at five. On synthetic data a simple rule is indistinguishable from the optimizer wherever the budget alone binds (the fixed rule at 5.2 against the optimizer's 4.4 errors under bandwidth limits, an interval spanning zero) and the admissibility filter rather than the optimizer does the work under misaligned reports (15.3 against 35.0). A derived delay threshold cut 95th-percentile latency from 1,562.3 s to 25.7 s, with the error difference against the policy spanning zero. Forged metadata erases the filter's protection; a reused conformal threshold drops coverage to 0.745 against a nominal 0.90; measured LTE traces inflate tail latency by up to 8.7 times; six uncoordinated controllers cost the exact knapsack 20.0 errors against 3.9 for a greedy ranking. On real data the ranking reverses and then proves fragile. Across seven event splits the optimizer leads the fixed rule in 15 of 21 comparisons on the damage label this paper chose, in 14 of 21 once class imbalance is handled everywhere and in 11 of 21 on a label the dataset's own annotators defined, where its lead under normal connectivity falls from all seven splits to three. A twenty-five-cell diagnostic attributes the text pathway's majority-class output to class imbalance rather than to the dimensionality of the reduction. No tested representation clears a bar fixed in advance. Transferred to real data the delay threshold keeps a 67- to 74-fold latency reduction in all seven splits while costing errors rather than saving them, because the real corpus does not decay evidence while a decision waits. What survives is narrow and we state it as such: the formulation earns its place by naming the quantities that matter and supplying a threshold, never by the solver, which did not reliably beat a greedy alternative. The direction of any real-data comparison is a property of the label as much as of the method.

***Index Terms*—edge AI, degraded connectivity, multimodal fusion, evidence admissibility, conformal prediction, adversarial metadata, disaster data**

I. Introduction

A flood, a wildfire or an earthquake produces satellite imagery, field photographs, public reports, weather measurements and infrastructure alarms inside the same hour and almost none of it arrives together. The image waits for the next satellite pass. The field report describes the neighbouring valley. The network that would carry the raw upload is the network the event has just damaged. A system that waits for everything can be accurate and useless at once, because the decision it supports was needed an hour ago.

This is an edge problem before it is a machine learning problem: which evidence deserves the remaining bandwidth, whether a late observation is worth waiting for and whether a given report belongs to this incident at all. The three decisions are coupled: waiting for an image consumes the time in which other evidence ages; admitting a report from the wrong event corrupts the fusion the budget was spent to enable.

This paper formulates the three decisions jointly and then measures whether the formulation earns its complexity. We treat the coupled problem as evidence budgeting across concurrent incident windows, solve it exactly and compare the resulting policy with the simple rules a practitioner would otherwise deploy. Those rules receive the same information as the optimizer: the same delay forecast, the same incident metadata and the same byte budget. Much of the apparent value of an adaptive method is the value of the information it is fed; the only way to separate the two is to give a simple rule the same information.

The answer is not the one we set out to demonstrate. On the synthetic benchmark at twenty seeds a simple rule derived from the formulation is indistinguishable from the optimizer the formulation justifies wherever the budget alone binds, while the gains elsewhere come from the delay threshold with the admissibility filter rather than from solving the program exactly. On real data the picture changes again, then proves fragile: against that same fixed rule the optimizer leads in 15 of 21 comparisons across seven event splits on the label this paper chose, in 11 of 21 on a label the dataset's own annotators defined, while on the chosen label a single-modality baseline that the formulation would never suggest beats both on errors, for a reason no term in the formulation represents. We report each result, identify its mechanism and draw the consequence.

A. What a Reader Gains

Because the solver is not where the value lies, it is fair to ask what this paper offers beyond "use simple rules and test them on your data". We claim four things. *First*, a formulation whose first-order condition (7) is a closed-form delay threshold; collapsed to a constant estimated on calibration data, it gives a wait rule needing no solver, cutting tail latency from 1,562.3 s to 25.7 s, with the error difference against the policy spanning zero. On real data with a constructed arrival delay it keeps that reduction in all seven event splits while costing 35.0 errors out of 968, so the pre-specified test of whether it carries over returns a negative answer (Section VIII-D). What the threshold buys is lower latency, at a cost that depends on whether the deployment's evidence goes stale while a decision waits. *Second*, a recipe for deriving three candidate rules from the same quantities, a byte-profile fixed rule, the constant threshold and a greedy ranking with a summary floor, with an information-matched protocol for choosing among them on held-out events; which rule wins changed between our two workloads while the derivation did not. *Third*, evidence about which assumptions fail: split-conformal coverage fails across operating conditions and, on real data, within a condition when calibration and test windows come from different disasters; the byte model is contradicted by measured payloads; the value function cannot detect a modality carrying no predictive signal for the label under test, a statement about this pathway on this corpus rather than about the modality; a twenty-five-cell diagnostic attributes the pathway's behaviour to class imbalance rather than to the representation; and the direction of the real-data comparison changes with the label, so a ranking measured on one labelling is not evidence about another (Section VIII-F). *Fourth*, two failure modes for contested deployments: metadata admissibility degrades almost linearly to nothing as forging capability grows, while the exact knapsack loses far more than a greedy rule when controllers cannot coordinate. A reproducible package regenerates every table and figure from frozen result files. The system is analyst decision support; it does not dispatch, order evacuation or allocate aid.

TABLE I
NOTATION.

Symbol	Definition
i, s	Incident window; slot of K windows sharing budget and link.
m	Evidence modality: image, text or sensor.
$p_{im}(y)$	Calibrated class probabilities of the representation sent.
$\hat{s}_i$	Edge estimate of incident severity.
r_{im}, q_{im}	Estimated source reliability and input quality, in $[0, 1]$.
f_{im}	Recency factor, $\exp(-\text{age}_{im}/\tau_m)$.
u_{im}, n_{im}	Local uncertainty and novelty of item (i, m).
$C_{im}(a)$, $L_{im}(a)$	Serialised byte cost and latency of action a.
$\hat{D}_{im}$	Forecast arrival delay of evidence not yet at the edge.
B, R	Per-window byte budget and link rate.
λ_b, λ_ℓ	Byte-cost weight and latency penalty per second.
$\hat{C}(x)$, $\hat{q}$	Conformal set and threshold at error level α.

II. Related Work and Positioning

Multimodal disaster intelligence. xBD supports building-damage assessment from satellite imagery [1]; FloodNet and Sen1Floods11 support aerial and radar flood understanding [2], [3], while CrisisMMD pairs text with images from seven disasters [4], where joint representations improve humanitarian-category prediction [5]. This literature defines what can be learned from disaster data, not what should be transmitted or withheld when the link degrades.

Edge–cloud collaboration. Edge computing moves computation toward the data [6], [7], collaborative inference partitions it by device and network state [8] and communication-efficient edge AI treats transmission as a first-order constraint [9]. Multimodal systems select what to transmit adaptively [10], [11], while AVERY adapts a vision–language model's split and compression to operator intent and bandwidth [12]. All of these optimize a single stream or device; none allocates one budget across concurrent incidents, asks whether a late observation is worth waiting for or checks that evidence belongs to the incident being decided.

Uncertainty, abstention and trust. Incomplete-modality learning addresses absent inputs [13], calibration makes probabilities interpretable [14]; selective prediction adds a reject option [15], while split conformal prediction gives marginal coverage under exchangeability [16], [17] that does not survive distribution shift [18], which is why we calibrate per condition and measure what reuse costs.

What is and is not claimed. No individual component is new; the combination is. Patent literature narrows the claim further: disaster-response systems over a disaster-chain knowledge base [19] and edge–cloud schemes combining evidential uncertainty, adaptive modality weights and temporal decay [20] predate this work, so we do not claim the first disaster-specific multimodal edge–cloud framework. The distinctive feature is methodological: the alternatives receive the same information and the outcome is reported even when it goes against the proposed method.

III. System Model and Problem Formulation

Table I collects the notation.

A. Incident Windows and Admissible Evidence

An incident window i has a grid cell c_i, a decision time t_i and a hazard tag h_i. Each evidence item m for that window carries its own cell, observation time and hazard tag; it is admissible only when all three are compatible:

$$\mathrm{adm}_{im} = \mathbf{1}\big[\|c_{im} - c_i\|_\infty \le 1\big]\,\mathbf{1}\big[|t_{im} - t_i| \le \tau_m^{\mathrm{adm}}\big]\,\mathbf{1}\big[h_{im} = h_i\big]. \quad (1)$$

An inadmissible item may only be held. In the reference configuration τ_m^{adm} is 7,200 s for imagery, 1,800 s for text and 600 s for sensor data; the cell tolerance is one grid step. The rule reads metadata only and never touches labels. Missing evidence is represented explicitly and never imputed.

B. Actions, Objective and Why This Is a Knapsack

For each admissible item the policy chooses one action $a \in \mathcal{A} = \{\mathrm{hold}, \mathrm{summary}, \mathrm{feature}, \mathrm{raw}\}$: a summary carries calibrated local probabilities and metadata, a feature carries a quantised embedding and the raw action carries the observation. Holding is not deletion; the item may remain in a local encrypted queue. The K windows of a slot share one budget KB and one link, so the slot policy solves

$$\begin{aligned} \max_{x} \; & \sum_{i \in s} \sum_{m} \sum_{a \in \mathcal{A}} x_{ima} \Big[v_{im}(a) - \lambda_b \tfrac{C_{im}(a)}{B} - \lambda_\ell L_{im}(a) \Big] \\ \text{s.t. } & \textstyle\sum_a x_{ima} = 1 \quad \forall (i,m), \quad \sum_{i,m,a} x_{ima}\, C_{im}(a) \le KB, \\ & x_{ima} \in \{0,1\}, \quad x_{im,\mathrm{hold}} = 1 \text{ if } \mathrm{adm}_{im} = 0. \end{aligned} \quad (2)$$

The latency of an action counts its transfer time on the shared link of rate R bytes per second and, for an item not yet at the edge, its forecast arrival delay:

$$L_{im}(a) = \frac{C_{im}(a)}{R} + \hat{D}_{im}\,\mathbf{1}[(i,m)\ \text{pending}], \quad L_{im}(\mathrm{hold}) = 0. \quad (3)$$

Holding costs no bytes and no latency. Queueing on the shared link is applied to every method at evaluation time and is not part of the per-item objective. Because each item contributes exactly one action and the slot shares a single budget, (2) is a multiple-choice knapsack [21].

Evidence value combines estimated severity, reliability, recency, an action-fidelity multiplier $\phi(a)$, local uncertainty, input quality and novelty against the admissible evidence. The exact form is the one the reference implementation evaluates:

$$v_{im}(a) = \phi(a)\ \hat{s}_i\ r_{im}\ f_{im}\ \big(\beta_0 + \beta_u u_{im} + \beta_q q_{im} + \beta_n n_{im}\big), \quad (4)$$

with $\phi = (0, 0.34, 0.64, 1.0)$ over (hold, summary, feature, raw) and $\beta_0 = 0.28$, $\beta_u = 0.34$, $\beta_q = 0.23$, $\beta_n = 0.35$, with $\lambda_b = 0.11$ in (2). Severity, reliability and recency enter multiplicatively, so any one near zero suppresses the item whatever the others say, while uncertainty, quality and novelty enter as a weighted sum over a positive floor β_0, so none of those three can suppress an item alone. Recency decays as $f_{im} = \exp(-\mathrm{age}_{im}/\tau_m)$, with τ_m of 3,600 s for imagery, 900 s for text and 300 s for sensor data. Local uncertainty is $u_{im} = 1 - \max_y p_{im}(y)$, a proxy for information gain rather than an estimate of it. Novelty is the Jensen–Shannon divergence between the item's local probabilities and the mean of the local probabilities of the admissible evidence already present in the window, that mean including the item itself:

$$n_{im} = \mathrm{JS}\Big(p_{im} \,\Big\|\, \bar{p}_i\Big), \qquad \bar{p}_i = \frac{1}{|P_i|} \sum_{m' \in P_i} p_{im'}, \quad (5)$$

where P_i is the set of admissible modalities whose local output has arrived. For an image that has not yet arrived no local output exists, so u_{im} is set to $1 - \max_y \bar{p}_i(y)$ over the evidence already present, q_{im} takes the prior mean $5/7$ and $n_{im} = 0$.

These coefficients are configuration rather than fitted quantities, set by hand as Section X records. They are stated here because the factor list alone is not enough to reproduce the policy from the paper. Two terms that earlier text named do not appear in (4): there is no disagreement term in the value function, while the action-fidelity multiplier is the whole of the action's effect on value. Disagreement is a separate per-window quantity, the largest Jensen–Shannon divergence between any two modalities selected in the window; it is zero when fewer than two are selected. It never enters the objective; it is what the conflict rule of Section VI-E thresholds.

C. The Wait-or-Act Decision

Because a pending item carries its forecast delay inside $L_{im}(a)$, waiting is not a heuristic bolted on beside the allocator. When the slot budget is not binding, (2) separates by item and a non-hold action a is selected for a pending item only when its net utility is positive:

$$v_{im}(a) - \lambda_b \frac{C_{im}(a)}{B} - \lambda_\ell \Big(\frac{C_{im}(a)}{R} + \hat{D}_{im} \Big) > 0. \quad (6)$$

Rearranging gives a threshold on the forecast delay,

$$\hat{D}_{im} < T_{im}(a) \;\equiv\; \frac{v_{im}(a) - \lambda_b\, C_{im}(a)/B}{\lambda_\ell} - \frac{C_{im}(a)}{R}, \quad (7)$$

and the policy waits for the item when $\max_{a \ne \mathrm{hold}} T_{im}(a) > \hat{D}_{im}$. The threshold rises with severity, reliability and the uncertainty of the evidence already in hand (all of which enter through $v_{im}(a)$) and falls with byte cost and with λ_ℓ. The coefficient λ_ℓ therefore states how many units of evidence value one second of delay costs and sweeping it traces an accuracy–latency frontier. When the budget binds, the knapsack trades waiting against other windows' needs. A decision that waits also ages the evidence it holds, which fusion accounts for through f_{im}.

Collapsing the threshold to a constant gives a rule that needs no optimizer: for the image-summary action, take

$$T^* = \mathrm{median}\{\, T_{im}(\mathrm{summary}) : (i,m) \in \mathcal{W}_{\mathrm{cal}} \,\}, \quad (8)$$

over calibration windows $\mathcal{W}_{\mathrm{cal}}$ whose image is pending, fix T^* before testing and wait for a late image only when $\hat{D}_{im} < T^*$. Across the five package seeds T^* was 196–219 s. Section VI-B compares that rule with the policy it came from.

D. Properties

Proposition 1 (Feasibility and exactness): Round each byte cost up to a multiple of $\delta = KB/2000$ and solve (2) by dynamic programming over the rounded budget [21]. Every returned allocation satisfies the true budget, because rounding up can only overstate cost; the returned allocation is optimal for the rounded problem. Each of the $3K$ items loses at most δ bytes of capacity to rounding, so the allocation is also optimal for the original problem whenever that problem has an optimum leaving at least $3K\delta$ bytes unused, which is 0.9% of the slot budget at $K = 6$.

Proposition 2 (Coverage within a condition): If calibration and test windows are exchangeable within an operating condition, the conformal set of (10) contains the true class with probability at least $1 - \alpha$ for that condition, where α is the conformal error level [16]. No guarantee is claimed across conditions; Section VI-D measures what is lost when one threshold is reused and Section VIII shows the premise failing within a condition on real data, where calibration and test windows come from different disasters.

Remark 1 (Severity cannot confirm): Severity appears only in the rule that routes a case to human review, never in the rule that assigns a confirmed status, so a severe but ambiguous case can be escalated but never confirmed. Severity is the quantity most likely to be correlated with operator anxiety and least likely to be well estimated.

IV. Method

Each edge node runs a lightweight model producing class probabilities calibrated by temperature scaling on calibration events [14]. The cloud holds one model per representation: a feature model trained only on the transmitted quantised embedding and a raw model trained on the full observation. No representation is scored with information it did not carry. Severity is estimated at the edge from the local probabilities of the admissible evidence already present.

For each window the 64 action combinations over three modalities are enumerated with their utilities and byte costs; the dynamic program processes windows in turn with a backpointer at every rounded budget level and the allocation is recovered by backtracking, at $O(K \cdot 64 \cdot 2000)$ operations per slot. All-hold is permitted only for a window with no admissible evidence.

Selected posteriors are combined by a weighted logarithmic opinion pool [22] over normalised weights,

$$p_i(y) \propto \prod_{m \in S_i} p_{im}(y)^{w_{im}}, \qquad \sum_{m \in S_i} w_{im} = 1, \tag{9}$$

where S_i is the set of modalities transmitted for window i. Because the weights are normalised, adding a corroborating modality changes the posterior through its content and not through the number of sources; corroboration is handled in the status rule instead. Before that normalisation the weight is $W_m q_{im} r_{im} f_{im}$, with W of 0.40 for imagery, 0.35 for text and 0.25 for sensor data. If a decision waits for an image, the evidence it holds ages by the waiting time before fusion.

TABLE II
Operational statuses. Thresholds are configuration parameters, not learned quantities.

Status	Rule	System behaviour
Human review required	$\hat{s} \geq 0.70$ and (set size $\neq 1$ or conflict)	Escalate with reason codes.
Insufficient evidence	Set size $\neq 1$	Withhold a categorical recommendation; queue to seek evidence.
Conflicting	Maximum pairwise JS divergence ≥ 0.15	Preserve competing evidence; queue.
Confirmed	Singleton set with confidence ≥ 0.90 from at least two modalities	Place high in the analyst queue; no autonomous dispatch.
Likely	Singleton set otherwise	Monitor; present uncertainty.

The fused posterior becomes a split-conformal set using the least-ambiguous set-valued classifier score [16]:

$$\hat{C}(x) = \{\, y : p_i(y) \geq \hat{q} \,\}, \tag{10}$$

where $\hat{q}$ is the $\lfloor \alpha(n_{\text{cal}} + 1) \rfloor$-th smallest true-class probability $p_j(y_j)$ over the n_{cal} calibration windows j processed under the same operating condition. When $\hat{q}$ exceeds 0.5 a set can contain at most one class, because two classes whose probabilities both reach the threshold would sum to more than one, so a non-singleton set is empty: no class reaches the threshold and the method abstains. Table II maps the set, the fused confidence and the conflict indicator to an operational status: a singleton set with confidence at least 0.90 from at least two modalities is confirmed, any other singleton likely; a non-singleton set is insufficient evidence; a maximum pairwise Jensen–Shannon divergence of at least 0.15 marks conflict; an estimated severity of at least 0.70 alongside either of those two escalates to human review.

Every decision is written to a provenance envelope recording the window, event and slot identifiers, each item's metadata, admissibility, action, bytes, reason code and source digest, the model, policy and calibration versions, the probabilities, the conformal set, the status and the rule that fired, plus a digest of the envelope itself. Completeness is computed by checking those fields for every selected item. A digest links a decision to retained source data; it does not establish that the source is true.

V. Reference Implementation and Evaluation Protocol

A. Implementation and Integrity Tests

The implementation is a set of deterministic Python modules called by the accompanying notebook. A configuration file fixes seeds, event splits, conditions, the byte model, recency scales, admissibility tolerances, policy coefficients and thresholds. Five integrity properties are tested rather than assumed: the transmission policy is unchanged when test labels and generating severities are permuted; budget-compliant methods never exceed a slot budget; the admissibility rule never rejects aligned evidence; the fixed rule with admissibility never sends

an inadmissible item; and the derived wait threshold is computed from calibration events only, is unchanged under label permutation and causes the rule to wait exactly for images whose forecast delay lies below it.

B. Workload and Network Model

A synthetic generator creates three classes and image, text and sensor feature vectors for 12 event identities with 180 windows each, every event carrying a grid location and a hazard tag; edge models see a prefix of each vector, the raw cloud model all of it and the feature model a quantised principal-component embedding. Byte sizes are assumptions: summaries 184–224 bytes, features 768–2,048 bytes, raw images 220–620 kB, raw text 350–1,800 bytes and raw sensor records 220–900 bytes. Section VIII measures these against real payloads and finds them wrong. Events D00–D05 train the models, D06–D07 calibrate them and D08–D11 are held out for testing, so no test event contributes to any fitted quantity. The severity estimator's mean training R^2 across seeds was 0.10, which is weak and is treated as such throughout.

Connectivity is modelled rather than packet-emulated; this is the most important limitation to keep in view. Windows form slots of $K = 6$ by decision time within an event. The windows in a slot share a budget of KB bytes and one link, served in order of estimated severity, so queueing delay applies to every method including the unconstrained reference. In the delayed-imagery condition 70% of images arrive after a log-normal delay with a 900 s median and the policy sees only a forecast with multiplicative log-normal error ($\sigma = 0.4$). There is no MAC layer, routing or contention model in the base configuration; what it captures is the coupling between byte volume, a shared serving order and decision latency. Section VII-B relaxes two of those assumptions by adding loss with retransmission and capacity replayed from third-party measurements.

C. Questions, Methods and Conditions

The confirmatory evaluation (Section VI) asks whether adaptive selection beats simple budget-compliant rules when bandwidth is the only constraint, how the wait-or-act decision trades errors against latency, whether admissibility keeps misaligned evidence out of fusion, how calibration and the review queue behave across conditions and whether exact optimization beats simple allocation when the shared budget binds. The waiting and admissibility questions are each answered against a rule that receives the same information as the optimizer. Questions asked after those answers were in hand (metadata forgery, link realism, uncoordinated controllers and a measured workload) are reported as exploratory in Sections VII and VIII.

Each seed provides 720 held-out test windows. We report the mean of twenty seeds with two-sided 95% t intervals and paired seed-level differences; these describe benchmark variability rather than population inference. The first five of the twenty are the seeds the original submission reported; they reproduce its means exactly, with the run stopping on any mismatch. Three of the eight paired error comparisons still include zero at twenty seeds, so we treat those three as inconclusive rather than as equivalence. The main protocol was not pre-registered. The information-matched baselines and the review-rule sensitivity analysis were added after an internal audit of the prototype; the wait threshold was fixed on calibration events and the review settings were left unchanged. The two contested-budget experiments of Section VI-F were pre-registered with frozen protocol files whose SHA-256 digests are checked at run time. Tables III and IV list the methods and conditions.

Among the general baselines only the proposed method applies the admissibility rule, so the whole-system comparison under misaligned evidence does not isolate it; the fixed rule with admissibility and the corresponding ablation do. Budget-compliant methods never exceeded a slot budget in any run

TABLE III
METHODS COMPARED. THE DERIVED THRESHOLD AND THE FIXED RULE WITH ADMISSIBILITY EXIST TO GIVE A SIMPLE RULE THE SAME INFORMATION THE OPTIMIZER RECEIVES.

Method	Description	Budget
Image only / text only	Local summary of one modality.	Compliant
Transmit all	Raw evidence for every available modality.	Reference; ignores budget
Confidence gate	Raw when local confidence < 0.72, else summary; downgraded until the window fits.	Compliant
Fixed compression	Feature for every available modality.	Compliant
Fixed rule	Image summary, raw text, raw sensor.	Compliant
Random within budget	Uniform choice among feasible action combinations.	Compliant
Wait if uncertain	Fixed rule, but wait for a late image only if text–sensor confidence < 0.90.	Compliant
Derived threshold	Fixed rule, but wait only if forecast delay $< T^*$ of (8).	Compliant
Fixed rule + admissibility	Fixed rule with the filter of (1).	Compliant
Proposed	Shared-budget knapsack (2) with admissibility and forecast delay.	Compliant

TABLE IV
OPERATING CONDITIONS.

Condition	Manipulation
Normal	10 Mbps link; 360 kB per-window budget.
Low bandwidth	256 kbps; 4,000-byte per-window budget.
Delayed imagery	2 Mbps; 70% of images late (median 900 s); forecast only.
Intermittent outage	512 kbps; 12 s interruption for 35% of windows.
Image / text missing	Remove all images / all text.
Noisy text	Soften text probabilities for 35% of windows; imperfect quality indicator.
Contradictory evidence	Point text toward a wrong class for 35% of windows; imperfect reliability indicator.
Misaligned evidence	Replace 30% of text items with reports from other events; 25% of those carry compatible metadata.

TABLE V

LOW BANDWIDTH (256 KBPS, 4,000 BYTES PER WINDOW). ERRORS OUT OF 720, MEAN OF TWENTY SEEDS WITH 95% INTERVALS. EVERY COLUMN, THE QUEUE SHARE INCLUDED, COMES FROM THE SAME TWENTY-SEED RUN. *IGNORES THE BUDGET.

Method	Errors [95% CI]	Bytes	p95 s	Queue
Image only	71.5 [59.1, 83.9]	224	0.27	7%
Text only	75.7 [63.9, 87.4]	196	0.26	6%
Transmit all*	4.3 [2.9, 5.7]	423,502	84.64	33%
Confidence gate	7.0 [5.0, 9.0]	1,063	0.47	30%
Fixed compression	4.4 [3.0, 5.8]	3,840	0.95	32%
Fixed rule	5.2 [3.6, 6.7]	1,860	0.59	34%
Random within budget	22.6 [18.5, 26.6]	1,695	0.60	27%
Proposed	4.4 [3.2, 5.6]	3,604	0.92	32%

and provenance completeness was 100% for every method and condition.

VI. CONFIRMATORY RESULTS ON THE SYNTHETIC BENCHMARK

A. When Bandwidth Is the Only Constraint, a Fixed Rule Is Enough

Table V gives the low-bandwidth condition. Among budget-compliant methods the policy and the fixed rule made statistically indistinguishable numbers of errors, a paired difference of -0.8 $[-1.7, 0.2]$. At five seeds the fixed rule held a nominal advantage of +0.2 errors; at twenty the nominal sign is the other way and the interval still spans zero, so what the wider run supports is that neither method is ahead here rather than that the simple rule wins. The policy spent 1,745 more bytes per window $[1{,}732, 1{,}757]$ and added 0.33 s to tail latency $[0.330, 0.334]$, buying image features that do not change the decision under this byte model. Transmit-all reaches 84.6 s at the tail because six windows share one link. The same pattern holds wherever the byte budget is the only binding constraint: under normal connectivity, under an intermittent outage and with imagery missing, the paired difference spans zero in each case. It does not hold once the evidence itself degrades. With text missing the policy is ahead by 2.75 errors $[1.03, 4.47]$, under noisy text by 1.30 $[0.13, 2.47]$ and under contradictory evidence by 2.50 $[0.80, 4.20]$, all three excluding zero at twenty seeds where five seeds could not separate them (Table XVI). When no decision involves waiting, admissibility or degraded evidence, the optimizer adds cost rather than value. Section VII-B shows that the cost grows once the link is modelled with loss and measured variability.

B. Waiting for Late Evidence: The Threshold Matters, the Solver Does Not

Table VI gives the delayed-imagery condition. Methods that always wait paid a median latency near 682.8 s. The policy sent 37% of images, those with short forecast delays or uncertain partial evidence as (7) predicts, holding its median latency at 0.19 s. Against the wait-if-uncertain rule, which ignores the forecast, it changed tail latency by $-1{,}197$ s $[-1{,}250, -1{,}145]$. Removing the latency penalty cost 9.1 errors $[5.6, 12.5]$ and made it wait for every image, which confirms that the term is doing the work (Table XVII).

TABLE VI

DELAYED IMAGERY. ERRORS OUT OF 720 WITH 95% INTERVALS, MEAN OF TWENTY SEEDS; MEDIAN AND 95TH-PERCENTILE MODELLED LATENCY; SHARE OF IMAGES SENT. *IGNORES THE BUDGET.

Method	Errors [95% CI]	Median s	p95 s	Image sent
Image only	71.5 [59.1, 83.9]	682.80	1,864.4	100%
Text only	75.7 [63.9, 87.4]	0.15	0.15	0%
Transmit all*	19.2 [15.5, 23.0]	688.69	1,869.5	100%
Confidence gate	25.6 [21.1, 30.0]	682.81	1,864.4	100%
Fixed compression	18.9 [14.9, 23.0]	682.84	1,864.4	100%
Fixed rule	23.4 [18.7, 28.1]	682.82	1,864.4	100%
Random within budget	32.1 [28.1, 36.2]	39.69	1,664.7	68%
Wait if uncertain	16.0 [12.3, 19.7]	42.56	1,562.3	53%
Derived threshold T^*	10.8 [8.6, 13.0]	0.18	25.7	31%
Proposed, $\lambda_\ell = 0.0004$	10.2 [8.3, 12.0]	0.19	364.9	37%

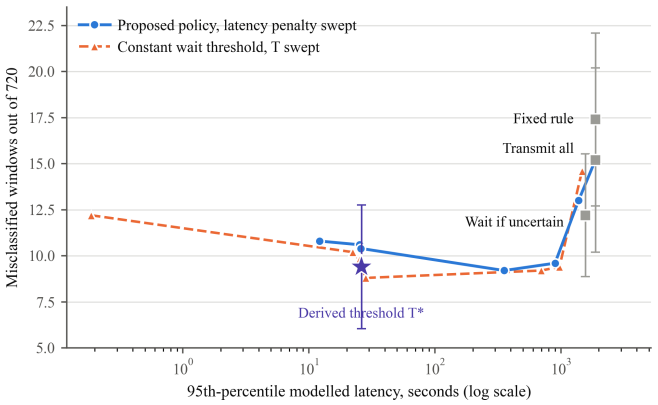


Fig. 1. Delayed imagery: errors against 95th-percentile modelled latency. The solid line traces the policy across latency-penalty values; the dashed line traces the constant-threshold rule across T. The star marks T^* derived from calibration events, not selected on test data. The sweep is the five-seed package run, not the twenty-seed rerun of Table VI.

The latency reduction does not require the optimizer. The derived-threshold rule sends the fixed rule's image summary whenever the forecast delay falls below the calibration-derived constant T^*. It sent 31% of images and made 10.8 errors at 25.7 s tail latency. Relative to it the policy changed errors by -0.7 $[-1.5, 0.2]$, tail latency by +339.2 s $[314.4, 363.9]$ and bytes by +688 $[670, 706]$. The error interval includes zero at twenty seeds as it did at five, which is the evidence for the claim in this subsection title. The policy's longer tail comes from per-window adaptivity: it waited more than 300 s for 5.9% of windows against 0.5% for the constant rule, when uncertain partial evidence raised the value of a late image. Those long waits bought nothing here. Fig. 1 shows that the constant-threshold frontier lies on or slightly below the policy's throughout.

The accuracy comparison carries a caveat that changes the conclusion. The policy changed errors by -13.3 $[-17.3, -9.2]$ against the fixed rule and by -5.9 $[-8.5, -3.2]$ against wait-if-uncertain. Those gains exist because the benchmark decays the weight of text and sensor evidence while a decision waits. With that decay removed, the forecast-ignoring baselines improve by 10.6 to 18.3 errors while the two forecast-using rules change by at most 0.5, so the ordering reverses: the fixed rule reaches 5.2

errors and wait-if-uncertain 5.4 against 9.7 for the policy and 10.7 for the derived threshold (sensitivity rows of the twenty-seed ablation results). Without staleness, both forecast-using rules trade 4.5 to 5.6 errors for their latency reduction. Whether acting early also improves accuracy therefore depends on how fast real evidence goes stale, which this benchmark assumes rather than measures. Section VIII-D tests exactly that on a corpus with no decay, where the trade is the one this sensitivity predicts.

C. Admissibility, Not Optimization, Keeps Foreign Reports Out

When 30% of text items came from other events, methods without the admissibility rule fused all of them. The proposed method fused 24% of misaligned items, exactly those whose metadata had been made compatible and rejected no aligned item. The complete system made 13.9 errors against 35.0 for the fixed rule without admissibility, a paired difference of -21.1 $[-26.2, -15.9]$. The fixed rule with the same filter made 15.3 errors; relative to it the proposed method changed errors by -1.5 $[-2.7, -0.3]$. The filter accounts for almost all of the gain and at twenty seeds the residual advantage of the optimizer over the filtered simple rule is small while its interval no longer spans zero. Within the proposed system, disabling admissibility added +16.2 errors $[11.1, 21.3]$; at five seeds that interval included zero, so the filter's own effect is now estimated precisely enough to read.

This test favours the filter by construction. Aligned evidence shares the window's cell and hazard tag exactly and is at most 60 s old, far inside the tolerances, while misaligned reports that were not spoofed carry the coordinates of another event on a 400×400 grid. Zero false rejections follow from the generator, not from the rule's robustness. Real metadata carries positional error, clock skew and missing geotags, none of which is represented here.

D. Conformal Coverage Does Not Survive a Change of Condition

Calibrating per condition kept empirical coverage between 0.889 and 0.899 at twenty seeds against a nominal 0.90; the shortfall is consistent with having only two calibration events. Reusing the normal-condition threshold is where the trouble starts. Coverage falls to 0.794 under contradictory evidence and to 0, from the five-seed package sweep, which has no twenty-seed rerun.745 with images missing, a loss of 13.8 percentage points in the condition where an operator is most likely to rely on the interval. Proposition 2 should not be stretched across conditions: a deployment that calibrates once at commissioning will be miscalibrated exactly when the network degrades. Section VIII shows that per-condition recalibration is necessary but not sufficient, because on real data the guarantee also fails across events.

E. The Review Queue Is the Real Operational Cost

Under low bandwidth about 30% of windows entered the combined abstention and review queue, roughly 61 queued windows per caught error; under contradictory evidence 53% did. The queue figures in this subsection come from the package's five-seed sensitivity run rather than the twenty seeds of Table V (the full table, for all nine conditions, is in the package). Under low bandwidth 98% of queued windows were correctly classified, so almost all reviewer effort confirmed a decision the system had already got right. Conformal abstention alone queued about 10% of windows and still caught every error; the pre-specified conflict rule nearly tripled the queue without catching an additional one. Across conditions the conflict rule added 11 to 43 percentage points of queue load for between zero and two additional caught errors per 720 windows. We keep the pre-specified settings rather than tune after the fact; the implication is that the conflict threshold is a staffing decision and should be set from reviewer capacity. The severity-based escalation rule was nearly inert, firing on 0.07% of windows against the 28% the generating severity would have escalated, because the severity estimator is weak.

TABLE VII
CONTESTED SHARED BUDGET (NOISE SCALE 1.75, TEN SEEDS): ERRORS OUT OF 720 BY PER-WINDOW BYTE BUDGET B. *IGNORES THE BUDGET; SHOWN AS A CEILING.

Method	700	900	1,200	1,500	1,900
Summaries only	78.6	78.7	78.6	78.6	78.8
Fixed rule, per window	78.6	75.1	68.9	65.6	64.0
Confidence gate	78.6	76.0	70.0	66.3	64.7
Random within budget	140.4	141.3	127.1	121.1	109.2
Greedy by uncertainty	76.0	71.2	68.9	63.3	61.9
Greedy, proposed utility	96.4	72.3	64.6	60.7	57.2
Proposed (exact knapsack)	96.8	72.6	64.9	59.7	57.1
Fixed rule, ignores budget*	57.6	57.5	57.7	57.5	57.4

F. When the Shared Budget Binds (Pre-registered)

None of the conditions above makes the shared budget bind, which is the only regime in which the knapsack can differ from a per-window rule. At the reference difficulty it could not have mattered: on calibration events, sending only summaries erred on 0.56% of windows against 0.44% for the unconstrained fixed rule. We therefore froze a protocol before running the proposed method on a harder condition, with the protocol file and its SHA-256 digest inside the package. Observation noise was scaled by 1.75, the smallest value in a pre-specified grid at which summaries alone err on at least 8% of calibration windows. Models were retrained at that difficulty, the link runs at 256 kbps with $K = 6$ and per-window budgets range from 700 to 1,900 bytes, with ten seeds. The pre-specified competitor is a shared-budget greedy rule that starts from summaries and upgrades items across the slot in order of local uncertainty; a greedy heuristic on the proposed utility isolates the exact solver. A difference counts only if its paired t-test survives Holm adjustment [23] across the five budgets at 0.05.

The optimizer did not beat the pre-specified competitor at any budget (Table VII, Fig. 2). At 700 bytes it was worse by +20.8 errors $[15.4, 26.2]$ with Holm $p < 0.001$. At the three largest budgets it led nominally by -4.0, -3.6 and -4.8

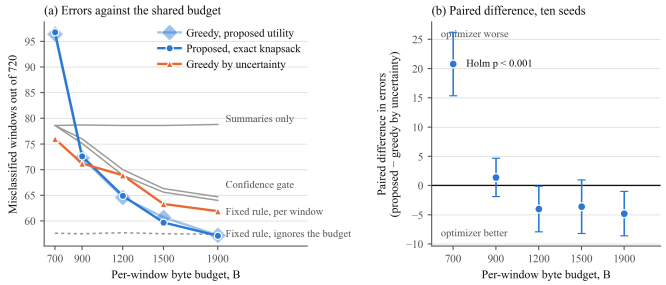


Fig. 2. Contested shared budget. (a) Errors against the per-window budget; marginal intervals are omitted because between-seed variation dominates them. The pale diamonds are the greedy heuristic on the proposed utility, which tracks the exact knapsack closely enough to be read as a halo around it. Random-within-budget is omitted because it runs from 140.4 to 109.2 errors; it is in Table VII. (b) The paired difference the analysis tests, with 95% intervals over ten seeds. Only the tight-budget failure survives Holm adjustment.

errors and none survived adjustment (Holm p 0.075 to 0.219). The exact solver matched a greedy solver on the same utility to within one error at every budget. One of those five solver-versus-greedy differences, at $B = 1{,}500$ bytes, survived Holm adjustment in the solver's favour ($p = 0.042$); it is 1.0 error in 720 (0.14% of windows). We do not regard that as a benefit worth an exact solver, but the pre-specified decision rule was met once and it would be wrong to say the solver never beat greedy.

The tight-budget failure has a clean mechanism. At 700 bytes the optimizer sent nothing for the image in 27% of windows and nothing for the text in 30% and held at least one modality in 47%, in order to buy raw sensor records elsewhere in the slot. It did so because the value function prices a held modality at zero rather than at the cost of losing it. Exact optimization does not repair a misspecified objective; it exploits it more efficiently than a greedy rule can.

A variant designed after these results adds a summary floor: every eligible item is sent at least as a summary and the knapsack chooses only upgrades. We froze a second protocol and tested it on ten new seeds. The floor removed the tight-budget failure. Its advantage over greedy-by-uncertainty replicated in direction at the three largest budgets (−3.0, −3.1 and −3.4 errors) but did not survive adjustment (Holm p 0.087 to 0.353) and was absent at the two tightest budgets (+0.2 and −0.9). The exact solver again matched a greedy solver with the same floor to within one error. The proposed value function may carry a benefit of roughly 0.4% of windows at moderate budgets; this benchmark does not establish it and the exact solver contributes nothing measurable.

VII. Exploratory Analyses

The experiments in this section were designed after the confirmatory results and are reported as robustness analyses, not as hypothesis tests. Each inherits the perturbation stream of the condition it modifies, so it is a controlled comparison rather than a differently drawn workload; we verified that at zero loss, with no trace and with one controller, every method's errors, bytes and latencies equal the earlier values to every printed digit.

TABLE VIII

Errors out of 720 as the share of misaligned reports carrying forged metadata rises, with an uncompromised text source (left) and a compromised source corrupting 35% of reports (right). Means of five seeds. The 25% column with an uncompromised source is the reference condition of Section VI-C and reproduces the five-seed misaligned column it was run against.

	Uncompromised source					Compromised source			
Method	0%	25%	50%	75%	100%	0%	25%	50%	100%
Fixed rule (no filter)	20.2	20.2	20.2	20.2	20.2	19.8	19.8	19.8	19.8
Fixed rule + adm.	7.0	9.4	11.4	15.0	20.2	10.6	14.0	16.0	19.8
Proposed	6.4	8.8	10.2	12.8	17.2	10.2	12.8	14.8	18.4

A. Adversarial Metadata: The Filter Is Exactly as Strong as Forgery Is Hard

The admissibility rule reads metadata and nothing else, so an adversary who can attach plausible location, time and hazard values to a report defeats it by construction. We hold the misaligned-evidence condition fixed and vary the fraction of misaligned reports whose metadata is set to the target incident's own values. Table VIII shows an almost linear collapse. With no forgery the filter is worth 7.0 errors against 20.2 for the unfiltered rule. With complete forgery it is worth nothing: 20.2 against 20.2. The admitted fraction tracks the forged fraction almost exactly (24.1%, 48.4% and 72.4% against 25%, 50% and 75%), so the rule admits the forged items and nothing else while rejecting no aligned item at any point. The optimizer contributes nothing to robustness: the proposed method tracks the filtered fixed rule within 0.4 to 3.0 errors across the range.

Two consequences follow. There is no safe operating point, so a deployment cannot argue that a low expected forgery rate makes the rule sufficient. And metadata admissibility is a cheap filter against accident (misrouted reports, stale caches, misconfigured clients), not a security control; where an adversary is in the threat model it must be backed by signed observations or attested sources, whose cost belongs in the byte budget. We have not evaluated such a scheme and the sweep models metadata substitution only. Crossing the forgery rate with a compromised text source that points 35% of reports toward a wrong class (the compromised-source half of that grid) shows that forgery alone is sufficient: with both capabilities the filtered rule reaches 19.8 errors, no worse than forgery alone.

B. Packet Loss and a Measured Link

The results so far treat the link as a constant-rate pipe. Two experiments relax that assumption. The first splits each payload into 1,500-byte packets, each lost independently with probability p and retransmitted, so the bytes placed on the wire grow by $1/(1-p)$ and each lost packet costs one 200 ms retransmission timeout. The expected number of lost packets grows with the payload, so the time penalty is super-linear in

TABLE IX
95TH-PERCENTILE MODELLED LATENCY IN SECONDS AGAINST PACKET LOSS, AT THE NORMAL PER-WINDOW BUDGET. *IGNORES THE BUDGET.

Method	0%	1%	2%	5%	10%
Fixed rule	0.09	0.10	0.10	0.12	0.14
Confidence gate	0.33	0.39	0.40	0.41	0.44
Fixed compression	0.10	0.11	0.12	0.14	0.17
Proposed	0.57	0.83	1.23	2.47	4.70
Transmit all*	2.24	2.89	3.64	6.12	10.72

bytes. Accuracy is unmoved, because nothing in this condition ties latency back to the decision; tail latency is not unmoved (Table IX). The fixed rule goes from 0.09 s to 0.14 s at 10% loss, a factor of 1.5; the proposed policy goes from 0.57 s to 4.70 s, a factor of 8.3. The gap between them widens from 6× to 34×. The optimizer's surplus bytes were close to free under an idealised link; under a lossy one they are the dominant term.

The second experiment replays measured cellular capacity traces in place of the constant rate [24]. Each trace records the milliseconds at which one MTU packet could be delivered on a real LTE uplink; we replay them unmodified, varying only the starting offset by slot and compare against a constant link at the same mean capacity, which isolates variability from speed (Table X). Holding mean capacity fixed, real variability inflates tail latency by up to 8.7×. On the Verizon driving trace, whose fifth-percentile throughput falls to 0.25 Mbps, the proposed policy's tail rises from 0.25 s to 2.17 s and the fixed rule's from 0.24 s to 1.49 s; the steadier traces inflate far less, so the penalty tracks the depth of the fades, not the average rate. Ranking the five methods by tail latency gives the same order under the replayed trace as under its mean-matched constant for all three traces, so the ordering survives the more realistic link; what does not survive is the idea that the optimizer's extra bytes are cheap.

C. Coordination Is Worth More to the Optimizer Than to the Simple Rule

Every result so far grants the shared-budget formulation its most favourable assumption: one controller owns all six windows in a slot. An incident area is more plausibly served by several edge nodes that cannot coordinate per slot over the degraded link they are conserving. We deal the slot's windows round-robin to C independent controllers, each allocating only its own windows against a share of the byte budget proportional to the windows it owns. The total budget is unchanged and all controllers still share the physical link. The contested-budget condition is used because it is the only regime in which the shared budget binds; $C = 1$ reproduces Table VII exactly.

The two per-window methods are controls: they allocate nothing across windows, so C cannot affect them and it does not (Table XI). The two coordinated methods separate sharply. Removing coordination costs the exact knapsack +20.0 errors $[14.2, 25.8]$ at $B = 900$ and the greedy uncertainty ranking only +3.9 $[0.8, 7.0]$. The penalty peaks at $B = 900$ rather than at $B = 700$ (+10.2), because at 700 bytes the centralised optimizer already fails badly enough that little is left to take away. At $C = 6$ the proposed method makes 92.6 errors against the greedy rule's 75.1, so a rough tie under centralisation becomes a clear loss once the assumption is dropped. The exact optimizer is not merely no better than a greedy ranking; it is more dependent on an architectural assumption that field deployments are least likely to satisfy and this benchmark charges nothing for the per-slot consensus it would need.

D. What the Policy Costs to Compute

We timed the allocation step on the contested-budget views as best-of-five wall clock on one workstation and recorded peak allocation (Table XII); no energy figure is inferred. The fixed rule decides a slot in 0.0012 ms; the exact knapsack takes 7.44 ms, about 6,190 times longer and holds 18 times the peak memory. Solve time grows faster than linearly up to four windows sharing a slot and close to linearly beyond six (1: 0.10 ms, 2: 0.42, 3: 0.88, 4: 1.44, 6: 2.37, 8: 3.24, 12: 4.83 ms), approaching the $O(K \cdot 64 \cdot 2000)$ bound once the fixed per-call overhead is amortised. The greedy heuristic on the proposed utility is slower than the exact program (15.5 against 7.44 ms) only because it is a Python loop while the dynamic program is vectorised.

VIII. A Reduced Comparison on Real Disaster Data

Every result above rests on a generated workload; the only measured inputs so far are the capacity traces. This section replaces the synthetic workload with one built from CrisisMMD [4]: 3,526 labelled report–image pairs from seven disasters in 2017, with real text, photographs, hazard types, arrival times recovered from the message identifiers, class imbalance and payload sizes. The policies are unchanged and the split is by event. The experiment was designed after the synthetic results, was not pre-registered and is exploratory.

A. Representations and Their Diagnostic

Both modalities are represented by CLIP embeddings [25], reduced by principal components fitted on training events only: eight components for text and ten for images, matching the dimensions the policy expects. Measured on the 968 held-out records before any budgeting decision, the image representation reaches 0.566 accuracy against a majority baseline of 0.490, while the text representation returns 0.490, the majority baseline to four decimal places: the text classifier predicts the majority class for every held-out record. A TF–IDF control in the same diagnostic reached the baseline in one of its six cells and fell below it in the other five. These are raw per-record accuracies, not the per-window error counts decided under a budget.

The first submission could not say whether that majority-class outcome was a property of the task, of the eight-component reduction or of class imbalance. A factor diagnostic now separates them. Twenty-five cells were scored on the rotation-0 split, each over five fitting seeds, drawn from five text representations (the eight shipped components, thirty-two

TABLE X
95TH-PERCENTILE MODELLED LATENCY IN SECONDS UNDER A REPLAYED MEASURED LTE TRACE AGAINST A CONSTANT LINK OF THE SAME MEAN CAPACITY. *IGNORES THE BUDGET.

	Verizon short			Verizon driving			T-Mobile driving		
Method	Const.	Replay	×	Const.	Replay	×	Const.	Replay	×
Fixed rule	0.24	0.37	1.5	0.24	1.49	6.2	0.23	0.25	1.1
Confidence gate	0.23	0.32	1.3	0.23	0.94	4.0	0.23	0.24	1.1
Fixed compression	0.26	0.42	1.6	0.25	2.20	8.7	0.23	0.27	1.1
Proposed	0.26	0.42	1.6	0.25	2.17	8.6	0.23	0.26	1.1
Transmit all*	3.84	7.22	1.9	3.67	20.42	5.6	1.34	3.74	2.8

TABLE XI
ERRORS OUT OF 720 AT $B = 900$ BYTES AS COORDINATION IS REMOVED. C IS THE NUMBER OF INDEPENDENT CONTROLLERS SHARING ONE SLOT. TEN SEEDS.

Method	$C = 1$	$C = 2$	$C = 3$	$C = 6$
Summaries only	78.7	78.7	78.7	78.7
Fixed rule, per window	75.1	75.1	75.1	75.1
Greedy by uncertainty	71.2	71.4	72.0	75.1
Proposed (exact knapsack)	72.6	78.3	81.5	92.6

TABLE XII
MEASURED DECISION COST AT $B = 1{,}200$ BYTES, PER SLOT AND PER WINDOW. EACH SEED IS TIMED FIVE TIMES AND THE FASTEST RUN KEPT; THE TABLE REPORTS THE MEAN OF FIVE SEEDS.

Method	Slot (ms)	Window (ms)	Peak (KiB)
Fixed rule	0.0012	0.0002	76
Fixed rule, per window	0.2070	0.0345	76
Greedy by uncertainty	0.4949	0.0825	180
Greedy, proposed utility	15.4829	2.5805	1,329
Proposed (exact knapsack)	7.4428	1.2405	1,329

components, the full 512-dimension CLIP embedding, a TF–IDF control and the deployed leading-four truncation), three class handlings [26] (none, balanced class weights, balanced resampling of the training rows) and three classifiers (the package edge model, logistic regression as a floor and the deployed truncation). The decision rule was fixed before the run: the pathway counts as informative under some tested representation only if the best cell clears the majority baseline by more than 0.02 accuracy and exceeds the majority classifier's macro-F1 in at least five of the seven rotations.

The rule returns false on the per-rotation conjunction the protocol records as primary, passing in none of the seven rotations. The best cell, the full CLIP embedding with balanced resampling, raises macro-F1 from the majority classifier's 0.219 to 0.452 on rotation 0 and its macro-F1 gain is positive in all seven rotations, yet its accuracy sits at 0.481 against a majority baseline of 0.490 and is below that baseline in every rotation. Three cells do clear the accuracy bar on rotation 0, by 0.028 to 0.046, all of them without class handling; none survives the rotations, passing in two, two and one of seven. So the answer is not that a better representation unlocks the pathway. What class handling recovers is macro-F1 rather than accuracy: the pathway carries class-discriminative information that an accuracy comparison on imbalanced classes hides, yet it still does not predict this label well enough to clear a bar set in advance. The claim this supports is therefore the original one with a sharper reason behind it: on this corpus the text pathway contributes no predictive signal that survives the test fixed in advance, the cause is class imbalance rather than the dimensionality of the reduction and nothing in the value function detects either.

B. What Carries Over

The comparison is reduced and the data forces the reductions. CrisisMMD has no sensor stream, so the workload has two modalities and the per-window action space shrinks from 64 combinations to 16, with a fixed-seed placeholder keeping the shared code path running and the runner asserting the sensor modality is never available. The annotations carry no geolocation, so the spatial term of (1) reduces to event identity. No severity field is independent of the damage label, so severity is held constant and its term is inert. Imagery is never missing, so the incomplete-modality machinery is not exercised. Three conditions carry over, not nine. Three methods are absent from Table XIII, including the fixed rule with admissibility, so the misaligned condition here cannot isolate the filter as Section VI-C does. CrisisMMD carries no delayed imagery, so the derived constant wait threshold, the clearest positive result of Section VI, was untested on real data in the first submission. It is now tested on a constructed delay drawn per window from the same log-normal family as the synthetic condition, the seed and the draw recorded in a protocol frozen before the run, with the protocol's digest in the run manifest. The delay is therefore declared rather than observed, the honest description of what the corpus allows, while every other quantity stays real; Section VIII-D reports the outcome. Five seeds vary model fitting only: the data, the split and the 968 test windows are identical across them and the test set is two disasters, 795 windows from Hurricane Irma with 173 from the Iraq–Iran earthquake. Intervals bound fitting jitter rather than sampling error, several being zero-width because the method is deterministic given the data. These are the package seeds, the same five every published number from here on averages. The reruns below reproduce Table XIII exactly before any new column is read.

C. Results

Against the fixed rule the ranking does not carry over (Table XIII). The optimizer wins under normal connectivity

TABLE XIII

ALL THREE REAL-DATA CONDITIONS. ERRORS OUT OF 968; THE MAJORITY-CLASS BASELINE IS 494. BYTES AND TAIL LATENCY ARE FOR LOW BANDWIDTH. *IGNORES THE BUDGET.

Method	Normal	Low bw.	Misaligned	Bytes	p95 s
Image only	420.0	420.0	420.0	224	0.27
Transmit all*	435.8	435.6	447.4	129,728	30.71
Fixed rule	464.2	465.4	473.4	342	0.29
Confidence gate	437.0	465.8	474.4	1,390	0.56
Random within budget	462.0	465.8	474.6	1,212	0.52
Greedy by uncertainty	466.8	466.8	476.0	2,166	0.63
Greedy, proposed utility	447.0	466.8	462.2	1,896	0.63
Proposed (exact knapsack)	447.0	466.8	462.2	1,896	0.63
Fixed compression	483.0	483.0	486.0	3,072	0.80
Text only	494.0	494.0	494.0	196	0.26

TABLE XIV

PROPOSED MINUS FIXED RULE IN ERRORS, BY EVENT SPLIT. NEGATIVE MEANS THE OPTIMIZER WINS. ROTATION 0 IS THE SPLIT OF TABLE XIII. COUNTS DIFFER BECAUSE THE EVENTS DIFFER IN SIZE.

Rot.	Test events	n	Normal	Low bw.	Misaligned
0	Irma + Iraq–Iran	968	−17.2	+1.4	−11.2
1	Harvey + Iraq–Iran	1,065	−3.0	−5.4	+2.8
2	Harvey + Maria	1,754	−0.2	+1.0	+1.6
3	California + Maria	1,393	−18.8	−12.8	+8.2
4	California + Mexico	709	−3.0	−1.0	+18.8
5	Mexico + Sri Lanka	273	−7.6	−8.6	−6.4
6	Irma + Sri Lanka	890	−57.2	−33.2	−47.6

by 17.2 errors [15.8, 18.6] and under misaligned evidence by 11.2 [7.8, 14.6] and loses only under low bandwidth, by 1.4 [0.7, 2.1], which is 0.14% of the test set. Reporting only the low-bandwidth column would have reproduced the synthetic conclusion by selection. One assignment of seven events to three splits produced that pattern and the test set is dominated by one hurricane, so we rotated the event list through seven pre-specified assignments, each keeping the same 3/2/2 shape, so that every event serves in the test set at least once (Table XIV). The rotations were fixed before running and none was discarded. The direction survives the rotation and the per-condition pattern does not. The optimizer is ahead of the fixed rule in 15 of the 21 comparisons and under normal connectivity in every one of the seven, but the loss under low bandwidth occurs in 2 rotations of 7 and the win under misaligned evidence in 3. What generalises from this corpus is therefore narrower than Table XIII alone suggests: on real data the optimizer is ahead of the fixed rule more often than not, decisively so when the link is not the constraint and which way any single condition falls depends on which disasters land in the test set.

The comparison that matters most for this paper is between the exact solver and a greedy heuristic on the same utility; on real data the two are identical: the same errors, bytes per window and 95th-percentile latency in all three conditions and at every seed, a paired difference of exactly 0.0. A greedy ranking by local uncertainty, which does not use the utility, makes 466.8 errors under normal connectivity against 447.0, so the utility is doing work even though the solver is not (Fig. 3).

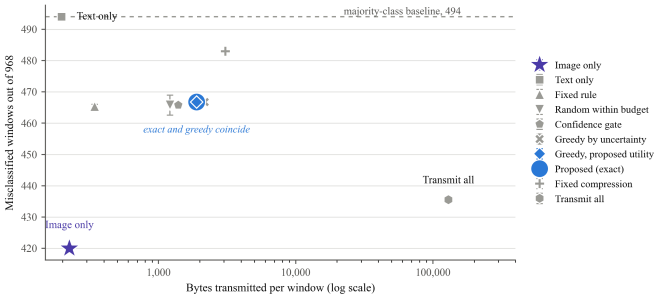


Fig. 3. Errors against bytes on real data under low bandwidth. Intervals are fitting jitter across five seeds on one fixed test set and are zero-width for the deterministic methods. The exact knapsack and the greedy heuristic over the same utility occupy the same point.

What beats everything on errors is image-only, at 420.0 in every condition, ahead of the optimizer by 46.8 errors [46.2, 47.4] under low bandwidth while sending 224 bytes. Its value cannot vary, because the conditions perturb text and bandwidth while image-only is sensitive to neither, so its constancy is not robustness; the same method makes 71.5 errors against the fixed rule's 5.2 on the synthetic benchmark, so a method near-worst on one workload and best on the other is telling us about the workloads. The reason is in the same table: text-only scores the majority-class count, 494.0, in every condition and everything fusing text pays for it. Severity, reliability, recency, fidelity, uncertainty, quality and disagreement are properties of the source or the signal; none asks whether this modality has ever predicted this label, which a held-out estimate of predictive contribution would. Transmit-all is also ahead on errors, by 31.2 [30.2, 32.2], while sending 129,728 bytes against 1,896 at 30.7 s of tail latency against 0.63 s.

Two assumptions do not survive the move. Conformal coverage, which held between 0.883 and 0.898 across the synthetic conditions, falls to 0.816 under normal connectivity and 0.745 under low bandwidth against a nominal 0.90. Calibration and test windows come from different disasters, so the exchangeability that Proposition 2 assumes fails within a condition; per-condition recalibration cannot repair that and the 7.2-point spread across the three conditions shows a condition-dependent component this explanation does not account for. The byte model of Section V-B is contradicted by measurement: real images run from 1,763 to 1,719,070 bytes with a median of 103,142.5 against an assumed 220,000 to 620,000, so the assumed range is both too narrow and centred too high and every report is smaller than the smallest summary the method defines, 39 to 181 bytes against an assumed 350 to 1,800. Sending a report verbatim is cheaper than summarising it, so the summarise-or-send decision is vacuous for that modality on this corpus. Byte sizes are measurable before a budget is designed around them.

D. The Constant Threshold on Real Data

The derived constant threshold is the clearest positive result of the synthetic benchmark, so it is the one worth testing

hardest. CrisisMMD carries no delayed imagery, so the arrival delay is constructed from the same log-normal family as the synthetic condition, drawn once from a protocol frozen before the run, with the protocol's digest in the run manifest; every other quantity stays as the corpus supplies it. T^* is computed on calibration events only, as (8) defines it, landing between 164.0 and 186.8 s across the rotations.

The latency half of the synthetic result carries over without qualification and the accuracy half reverses. On rotation 0 the threshold rule holds 95th-percentile latency at 25.4 s where the always-wait rules reach 1,833.9 s, a factor of 72. It holds that latency in all seven rotations: 25.1 to 26.8 s against 1,739.2 to 1,914.8 s. It pays for that with errors, 486.0 against 451.0 out of 968. The decision rule for this test was fixed in advance and asks for both halves: the threshold rule's tail latency below wait-if-uncertain's in all seven rotations, which holds, together with its error count inside the proposed policy's paired interval in at least five, which holds in one. The rule therefore returns *does not carry over*. The reason is specific rather than general. Acting early improved accuracy on the synthetic benchmark only because that benchmark decays the weight of evidence while a decision waits, as Section VI-B already reports. The real corpus has no such decay, so waiting costs nothing in evidence quality and the latency saved is bought with accuracy rather than alongside it.

One observation bears on baseline selection. The fixed rule and wait-if-uncertain are indistinguishable here at every seed and rotation because wait-if-uncertain's branch, text and sensor confidence below 0.90, which on this two-modality corpus reduces to text confidence alone, fires on every window of this corpus, so the baseline meant to ignore the forecast collapses into the one that always waits. A baseline whose branch always fires is not a comparison.

E. Class Imbalance in the Budgeting Pipeline

Section VIII-A attributes the text pathway's behaviour to class imbalance, so the question is what the budgeting comparison does when the imbalance is handled everywhere rather than only diagnosed. Balanced resampling [26], the handling the diagnostic selected, was applied to every per-modality edge and cloud model before the comparison was rerun. The unhandled column reproduces all ten methods of Table XIII exactly, which is the precondition for reading the handled column.

The direction of the headline claim survives and its magnitude does not. The optimizer stays ahead of the fixed rule in 14 of the 21 comparisons against 15 without handling, together with all seven rotations under normal connectivity in both columns. The sizes move far more than that count suggests: under normal connectivity on rotation 0 the gap widens from −17.2 errors to −112.6. Methods also move in opposite directions. The optimizer improves from 447.0 to 385.6 errors and transmit-all from 435.8 to 377.8, while every single-modality rule gets worse, text-only from 494.0 to 576.4 and image-only from 420.0 to 466.6. Handling the imbalance makes the fused prediction better and the single-modality prediction worse, which is consistent with resampling trading majority-class accuracy for minority-class recall. The comparison is thus sensitive to class handling in size while not in sign. The single-modality rule that beats everything in Table XIII does so partly because the imbalance was left in place.

TABLE XV

THE EXTERNALLY DEFINED LABEL. PROPOSED MINUS FIXED RULE IN ERRORS BY ROTATION, NEGATIVE MEANS THE OPTIMIZER WINS, ON THE THREE MOST FREQUENT CRISISMMD HUMANITARIAN CATEGORIES. CURRENT IS THE DEPLOYED PIPELINE; HANDLED APPLIES THE BALANCED RESAMPLING OF SECTION VIII-E. DISCARDED CATEGORIES: NOT HUMANITARIAN (398 RECORDS), INJURED OR DEAD PEOPLE (187), AFFECTED INDIVIDUALS (150), VEHICLE DAMAGE (21), MISSING OR FOUND PEOPLE (7).

	Normal		Low bw.		Misaligned	
Rot.	Curr.	Hand.	Curr.	Hand.	Curr.	Hand.
0	−8.0	+0.4	−1.8	+1.2	−3.8	+23.2
1	−10.4	+10.0	−13.2	+17.0	−15.8	+18.2
2	+9.8	−21.0	−27.0	−24.0	−57.8	−14.8
3	+7.4	−7.0	+14.2	+11.2	−4.4	+9.8
4	+19.6	+100.6	+8.6	+119.8	−3.8	+102.0
5	0.0	−3.4	+1.0	+1.4	+1.4	+1.2
6	−3.0	−4.2	+12.0	−22.0	+4.8	−13.6
Ahead	3/7	4/7	3/7	2/7	5/7	2/7

F. A Label the Paper Did Not Choose

Every real-data number above scores the damage label annotated from the photograph, which is this paper's own choice, so the sharpest available test is to replace it with one the dataset's annotators defined. CrisisMMD labels every record with a humanitarian category independently of the damage field [4]. The decision machinery is fixed at three classes, so the task is restricted to the three most frequent, keeping 2,763 of 3,526 records in three categories: other relevant information; infrastructure and utility damage; rescue, volunteering or donation effort. That restriction is a property of the machinery rather than a selection made after seeing results. The caption of Table XV lists the five discarded categories. This is not the published eight-category task and no published baseline is reproduced here; Section X says why neither was possible.

This is where the comparison fails. On the annotators' label the optimizer is ahead of the fixed rule in 11 of the 21 comparisons rather than 15, together with 3 of the 7 rotations under normal connectivity rather than all 7. With the class handling of Section VIII-E applied as well it is ahead in 8 of 21. The claim in Section VIII-C that survives the event rotation, that the optimizer is ahead under normal connectivity in every one of the seven, does not survive a change of label on the same records, the same splits and the same conditions. What generalises from this corpus is therefore narrower again than the rotation analysis alone establishes: the direction of the real-data result is a property of the label as much as of the method, so a deployment cannot read Table XIII as evidence that the optimizer will lead on its own labelling.

IX. What the Evidence Supports

One conclusion holds across both workloads: the value of this formulation is not in its solver. On real data a greedy heuristic over the proposed utility returned the exact program's allocation at every seed, a paired difference of exactly zero. At twenty seeds the optimizer is no longer beaten by the fixed rule anywhere on the synthetic benchmark, being indistinguishable from the fixed rule wherever the budget alone binds while leading it where waiting, admissibility or degraded evidence is in play. That corrects the five-seed reading this paper previously gave without rescuing the solver, because the gain comes from the threshold rule with the admissibility filter, both of which the formulation supplies with no solver at all. It earns its place by identifying which quantities matter (the forecast delay, the value of evidence already present, the incident metadata) while supplying the threshold (7) that turns them into a rule. What does not hold across both workloads is the ranking. On real data the optimizer leads the fixed rule in 15 of 21 comparisons on the label this paper chose, 14 of 21 once class imbalance is handled everywhere and 11 of 21 on the annotators' own label, where its lead under normal connectivity falls from all seven splits to three. Which simple rule wins is therefore a property of the workload, of the class handling and of the labelling, so the rule has to be derived and tested against held-out data from the deployment, labelled the way the deployment labels it. Seven checks follow, all within the simulation scope of this study. *One*, derive the candidates, a byte-profile fixed rule, the constant threshold (8) and a greedy ranking with a summary floor where a shared budget binds, then choose among them on held-out events, keeping the exact optimizer only if it beats all three. *Two*, before optimizing a value function check what it prices at zero; the tight-budget failure was a misspecification that exact optimization exploited. *Three*, treat metadata admissibility as protection against accident rather than adversaries, its value falling linearly with forging capability. *Four*, recalibrate conformal thresholds per operating condition and across the event boundary the deployment will cross, sizing the review queue from reviewer capacity. *Five*, check whether a shared-budget method still helps when controllers cannot coordinate; the exact knapsack lost far more than a greedy ranking. *Six*, add a held-out estimate of predictive contribution to the value function, since no existing term detects a modality carrying no signal for the label. Handle class imbalance in every per-modality model before reading the comparison: doing so left the direction of the real-data result intact while moving one rotation's gap from -17.2 to -112.6 errors and reversing which methods benefit. *Seven*, score the comparison on more than one labelling of the same records; the result here held across seven event splits yet not across a change of label, so an event rotation alone does not establish that a ranking will transfer.

The value of the wait-or-act decision itself rests on two quantities this benchmark assumes, how quickly an early decision loses value and how quickly evidence goes stale, which have to come from emergency-management partners and measured networks.

X. Limitations and Threats to Validity

The verification data are synthetic and well separated, which explains error rates below 1% for the stronger methods and makes the conformal sets nearly binary. Byte sizes, delay distributions, forecast errors and the rate at which evidence ages are assumptions on which the conclusions about fixed rules and waiting depend; Section VI-B shows the accuracy result under delay disappearing when evidence does not age. Section VIII removes a smaller part of this limitation than "real data" suggests: the link, budgets, loss model, slot structure, sensor modality and severity stay synthetic or fixed, six of nine conditions and three of eleven methods do not carry over and the arrival delay used to test the wait threshold on that corpus is constructed rather than observed. The text pathway's majority-class outcome is no longer unresolved: a twenty-five-cell diagnostic attributes it to class imbalance rather than to the dimensionality of the reduction, with no tested representation clearing a bar fixed in advance in more than two of seven splits.

The link model is the sharpest limitation for an edge-systems audience. Loss, retransmission and replayed measured capacity narrow it, but there is no protocol stack: no MAC layer, routing, contention between nodes, congestion control or bidirectional traffic. There is no hardware testbed and no energy measurement; Section VII-D reports wall-clock time and peak allocation on one workstation and we decline to infer joules from them. Section VII-C relaxes the single-controller assumption by static partitioning only. Every statement in this paper about edge deployment is therefore bounded by simulation on a workstation.

Policy coefficients and thresholds were set by hand; the latency penalty is accompanied by a sweep and the main protocol was not pre-registered. The admissibility test favours the filter by construction. The contested-budget difficulty was chosen by a rule fixed in advance, but the budget grid and the competitor were chosen by the authors and both protocols are timestamped only within the package. The severity estimator is weak, so severity-dependent behaviour is barely exercised. Twenty seeds and four test events per seed still limit statistical resolution: three of the eight paired error comparisons include zero, so we treat those three as inconclusive rather than as equivalence. Quadrupling the seeds reversed no reported direction while changing one nominal sign, the low-bandwidth comparison, against this paper's own earlier reading. On the real-data side the seeds vary model fitting only, which is why we rotate the event split; the seven assignments reuse the same events and are not independent. Social-media and sensor sources carry participation and placement bias. Improved disaster outcomes cannot be claimed without measured network traces, more disasters and prospective studies with emergency-management partners.

Two tests could not be completed on the available inputs, so we state the grounds rather than substitute something weaker in

silence. Reproducing the published CrisisMMD baselines [4] on the official split needs the published values and the image files, of which only derived embeddings remain. The split itself was obtained and holds the same 3,526 damage-task records used here, divided 2,468/529/529 rather than by event. No published multimodal baseline covers this task: the baselines of [5] exclude damage severity because it was annotated on images alone. The established tasks also cannot run at their published cardinality [4], two classes for informative against not-informative and eight for the humanitarian categories, because the reference implementation is fixed at three and is carried unchanged so its published numbers stay checkable. The three-category restriction of Section VIII-F is what that permits; it is not the published task and no reproduction gap is reported. An optional emulator run was not executed because the link-shell tooling is absent, so nothing here rests on packet emulation.

XI. Reproducibility, Oversight and Data Availability

The package contains the source, an environment lock, the configuration, split identifiers, calibration temperatures, thresholds, seeds, integrity tests, record-level predictions for every method and condition, provenance envelopes, the two frozen contested-budget protocols with their SHA-256 digests, plus scripts that regenerate every table and figure. On one CPU architecture reruns reproduce every output file byte for byte across Python 3.12 to 3.14. Across architectures about 10% of source observation vectors differ in their last bits, because the generator's Beta and Dirichlet draws reach platform libm; no reported quantity is affected, being written at six significant figures or fewer, with the provenance digest taken over the observation rounded to nine decimals. The real-data pipeline depends on a pretrained encoder and on PyTorch, whose versions are not pinned, so byte-for-byte reproduction of the derived records is not claimed; a digest of those records, the encoder identifier, the split, the label mapping and the source URL are recorded instead. The package changelog lists design errors found and corrected during development; results from before those corrections are superseded.

The method is decision support, not an autonomous responder. Reviewers receive the evidence set, the uncertainty, the conflict indicators, the reason codes and the status rule that fired; the interface must distinguish observed evidence from model-derived fields. Governance follows the NIST AI RMF lifecycle [27]; deployment requires agency authority, privacy and cybersecurity review, periodic recalibration (Section VI-D) and a staffing plan for the queue of Section VI-E. Raw transmission must be policy-gated.

The package is maintained at https://github.com/bdata-arindam/Edge at version 2.6.17, the version behind every result reported here, whose `SHA256SUMS` records a digest for each of its 113 files. The synthetic verification contains no personal data. Section VIII uses CrisisMMD v2.0, obtained from its authors under their terms and not redistributed; we ship derived features, byte counts and labels, not the posts or images. Operational use of field reports or agency data would require institutional, legal and privacy review.

TABLE XVI
Proposed (P), fixed rule (FR) and transmit all (TA) across every synthetic condition, mean of twenty seeds. Errors out of 720; bytes per window; 95th-percentile modelled latency in seconds. Every row comes from the same twenty-seed run, so this table shares its seed basis with Tables V and VI.

Condition	Err P	Err FR	Err TA	Bytes P	Bytes FR	p95 P	p95 FR
Normal	4.6	5.2	4.4	50,928	1,860	0.60	0.09
Low bandwidth	4.4	5.2	4.3	3,604	1,860	0.92	0.59
Delayed imagery	10.2	23.4	19.2	2,393	1,860	364.85	1,864.41
Outage	4.5	5.2	4.4	3,684	1,860	13.87	13.71
Image missing	14.1	14.1	14.1	1,636	1,636	0.19	0.19
Text missing	17.2	20.0	17.6	2,608	784	0.21	0.16
Noisy text	6.5	7.8	7.0	3,684	1,860	0.24	0.19
Contradictory	13.8	16.3	14.1	3,684	1,860	0.24	0.19
Misaligned	13.9	35.0	30.8	3,448	1,860	0.23	0.19

XII. Conclusion

We formulated disaster evidence transmission as a shared-budget knapsack across concurrent incidents in which waiting for late evidence is an explicit decision with a delay threshold and admissibility is checked before any evidence is valued or fused. On the synthetic benchmark at twenty seeds the simple rules the formulation generates are indistinguishable from the optimizer wherever the budget alone binds; a pre-registered test found the optimizer worse than its competitor at the tightest budget while one of its five differences survived adjustment in the solver's favour, by one error in 720; the admissibility filter's protection falls to nothing in proportion to an adversary's ability to forge metadata; and the conformal guarantee does not transfer across conditions. On real data the optimizer's lead over the fixed rule falls from 15 of 21 comparisons on the damage label this paper chose to 14 of 21 once class imbalance is handled and to 11 of 21 on the annotators' own. Under normal connectivity it falls from all seven event splits to three. A single-modality rule beats everything on that chosen label, partly because the imbalance was left in place. The text pathway is diagnosed: its majority-class behaviour is a class-imbalance effect rather than an artefact of the reduction, no tested representation clears a bar fixed in advance and the value function detects none of it. The delay threshold keeps a 67- to 74-fold tail-latency reduction in every split while costing accuracy rather than saving it, because that corpus does not decay evidence while a decision waits. What survives both workloads is that the formulation earns its place by naming the quantities that matter and supplying the threshold that turns them into a rule, never by the solver. Which simple rule to deploy has to be derived against the data a system will see and scored on the labelling that system will use. Field value under a real protocol stack and on edge hardware remains to be established.

TABLE XVII

COMPONENT ABLATIONS AT TWENTY SEEDS (COMPONENT REMOVED FROM POLICY AND FUSION; POSITIVE MEANS IT HELPED; SEVERITY REPLACED BY ITS TRAINING MEAN). SEVEN OF THE ELEVEN ABLATIONS ARE DISTINGUISHABLE FROM ZERO AT TWENTY SEEDS AGAINST THREE OF ELEVEN AT FIVE; THE FOUR EXACT ZEROS ARE COMPONENTS THE CONDITION DOES NOT EXERCISE. ROWS ARE ORDERED BY THE MAGNITUDE OF THE EFFECT.

Component (condition)	Δ errors, ablated minus full [95% CI]
Severity (low bandwidth)	+0.00 [0.00, 0.00]
Severity (normal)	+0.00 [0.00, 0.00]
Novelty (contradictory)	+0.00 [0.00, 0.00]
Novelty (noisy text)	+0.00 [0.00, 0.00]
Reliability (noisy text)	−0.45 [−0.89, −0.01]
Recency (delayed)	−0.50 [−0.99, −0.01]
Reliability (contradictory)	+2.75 [1.71, 3.79]
Quality (contradictory)	+2.80 [1.52, 4.08]
Quality (noisy text)	+3.45 [2.35, 4.55]
Latency penalty (delayed)	+9.05 [5.64, 12.46]
Admissibility (misaligned)	+16.20 [11.07, 21.33]

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